

US3M

1000 V, 3 A ultrafast recovery rectifier in SMC

20 January 2025 **Product data sheet**

1. General description

Ultrafast recovery rectifier, encapsulated in an SMC package.

2. Features and benefits

- Reverse voltage: $V_R \leq 1000$ V
- Forward current: $I_F \leq 3$ A
- Ultrafast recovery time $t_{rr} \leq 75$ ns
- Pt doped life time control
- Ideal for automated placement
- Glass passivated chip junction
- High forward surge capability

3. Applications

- Rectification
- Reverse polarity protection
- Fast switching
- Freewheeling applications

4. Quick reference data

Table 1. Quick reference data

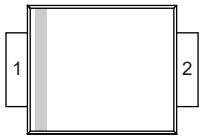
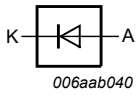
Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$I_{F(AV)}$	average forward current	$\delta = 0.5$; $f = 20$ kHz; square wave; $T_{sp} \leq 131$ °C		-	-	3	A
V_{RRM}	repetitive peak reverse voltage	$T_j = 25$ °C		-	-	1000	V
V_R	reverse voltage			-	-	1000	V
V_F	forward voltage	$I_F = 3$ A; pulsed; $T_j = 25$ °C	[1]	-	-	1.7	V
		$I_F = 3$ A; pulsed; $T_j = 125$ °C	[1]	-	1.12	-	V
I_R	reverse current	$V_R = 1000$ V; pulsed; $T_j = 25$ °C	[1]	-	-	10	µA
		$V_R = 1000$ V; pulsed; $T_j = 125$ °C	[1]	-	-	350	µA

[1] Very short pulse, in order to maintain a stable junction temperature.

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5. Pinning information

Table 2. Pinning information

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	K	cathode	 SMC (SOD1003-1)	
2	A	anode		

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
US3M	SMC	plastic, surface mounted package; 2 terminals; 6.86 mm x 6.11 mm x 2.34 mm body	SOD1003-1

7. Marking

Table 4. Marking codes

Type number	Marking code
US3M	AYB3

8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).

Symbol	Parameter	Conditions		Min	Max	Unit
V_{RRM}	repetitive peak reverse voltage	T_j = 25 °C		-	1000	V
V_R	reverse voltage			-	1000	V
V_{RMS}	RMS voltage			-	700	V
I_F	forward current	δ = 1; T_{sp} ≤ 126 °C		-	4.2	A
$I_{F(AV)}$	average forward current	δ = 0.5; f = 20 kHz; square wave; T_{sp} ≤ 131 °C		-	3	A
I_{FSM}	non-repetitive peak forward current	t_p = 8.3 ms; single half sine wave (applied at rated load condition); $T_{j(init)}$ = 25 °C		-	100	A
P_{tot}	total power dissipation	T_{amb} ≤ 25 °C	[1]	-	0.92	W
			[2]	-	1.25	W
T_j	junction temperature			-55	150	°C
T_{stg}	storage temperature			-55	150	°C

[1] Device mounted on an FR4 PCB, single-sided copper, tin-plated and standard footprint.
[2] Device mounted on an FR4 PCB, single-sided copper, tin-plated, mounting pad for cathode 1 cm^2.

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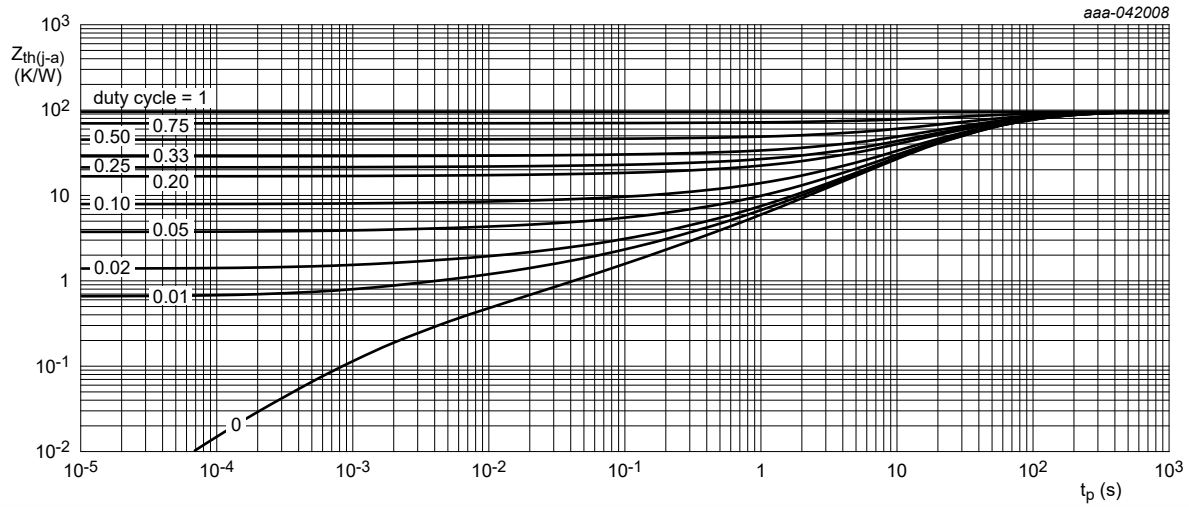
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9. Thermal characteristics

Table 6. Thermal characteristics

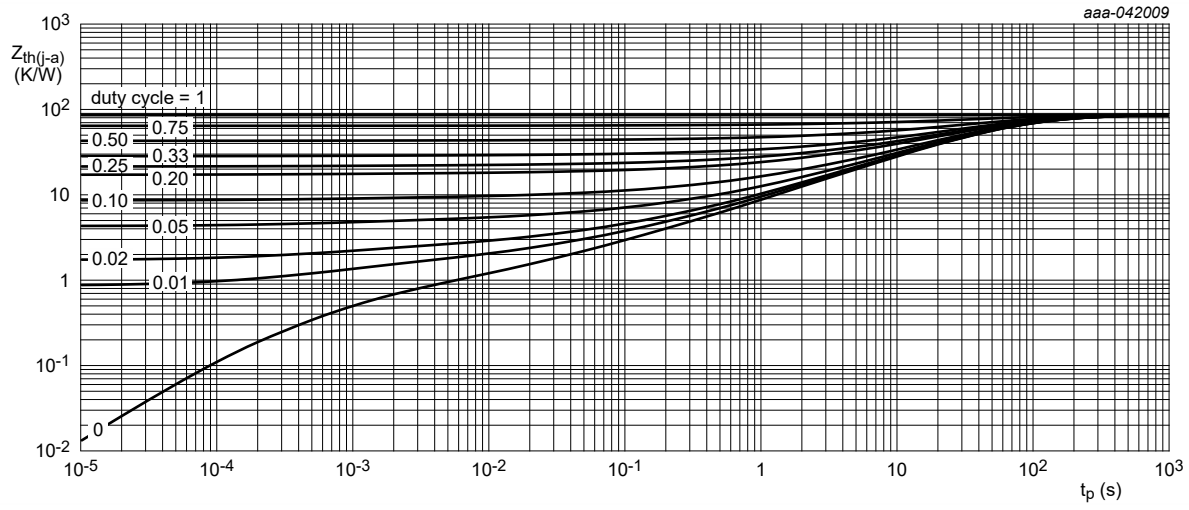
Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1]	-	-	135	K/W
			[2]	-	-	100	K/W
$R_{th(j-sp)}$	thermal resistance from junction to solder point		[3]	-	-	15	K/W

[1] Device mounted on an FR4 PCB, single-sided copper, tin-plated and standard footprint.
[2] Device mounted on an FR4 PCB, single-sided copper, tin-plated, mounting pad for cathode 1 cm^2.
[3] Soldering point of cathode tab.



FR4 PCB, standard footprint

Fig. 1. Transient thermal impedance from junction to ambient as a function of pulse duration; typical values



FR4 PCB, mounting pad for cathode 1 cm^2

Fig. 2. Transient thermal impedance from junction to ambient as a function of pulse duration; typical values

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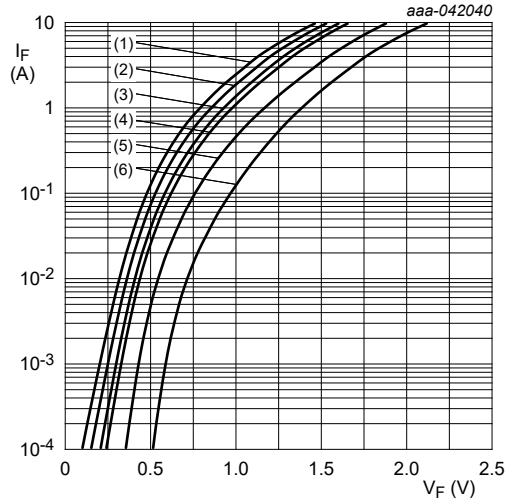
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10. Characteristics

Table 7. Characteristics

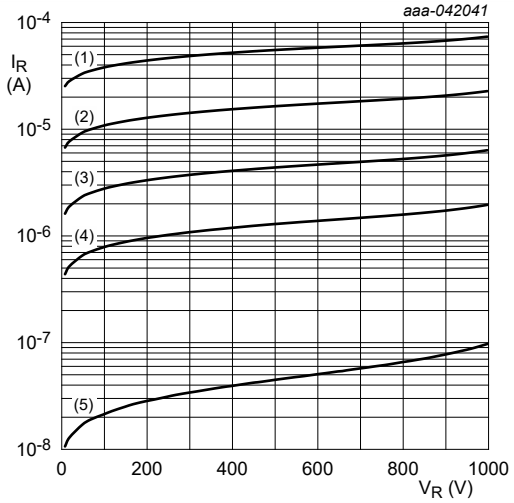
Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$V_{(BR)R}$	reverse breakdown voltage	I_R = 100 µA; pulsed; T_j = 25 °C	[1]	1000	-	-	V
V_F	forward voltage	I_F = 3 A; pulsed; T_j = 25 °C	[1]	-	-	1.7	V
		I_F = 3 A; pulsed; T_j = 125 °C	[1]	-	1.12	-	V
I_R	reverse current	V_R = 1000 V; pulsed; T_j = 25 °C	[1]	-	-	10	µA
		V_R = 1000 V; pulsed; T_j = 125 °C	[1]	-	-	350	µA
C_d	diode capacitance	V_R = 4 V; f = 1 MHz; T_j = 25 °C		-	23	-	pF
t_{rr}	reverse recovery time ; step recovery	I_F = 0.5 A; I_R = 1 A; $I_{R(meas)}$ = 0.25 A; T_j = 25 °C		-	52	75	ns

[1] Very short pulse, in order to maintain a stable junction temperature.



pulsed condition
(1) T_j = 150 °C
(2) T_j = 125 °C
(3) T_j = 100 °C
(4) T_j = 85 °C
(5) T_j = 25 °C
(6) T_j = -40 °C

Fig. 3. Forward current as a function of forward voltage; typical values

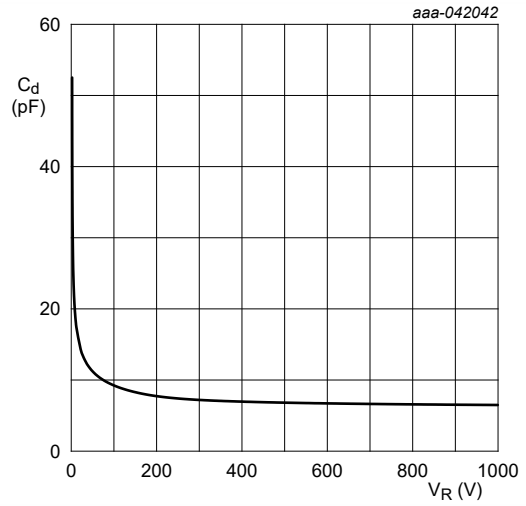


pulsed condition
(1) T_j = 150 °C
(2) T_j = 125 °C
(3) T_j = 100 °C
(4) T_j = 85 °C
(5) T_j = 25 °C

Fig. 4. Reverse current as a function of reverse voltage; typical values

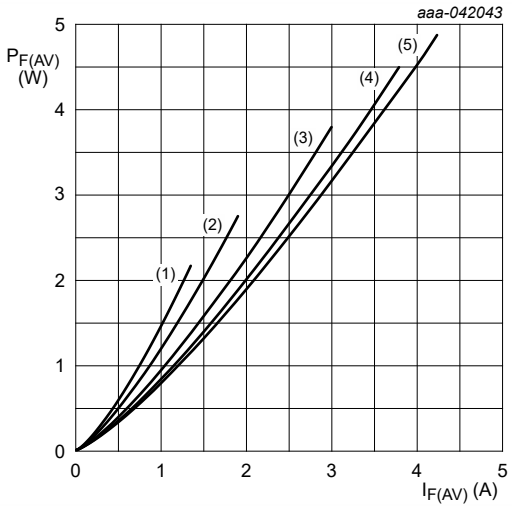
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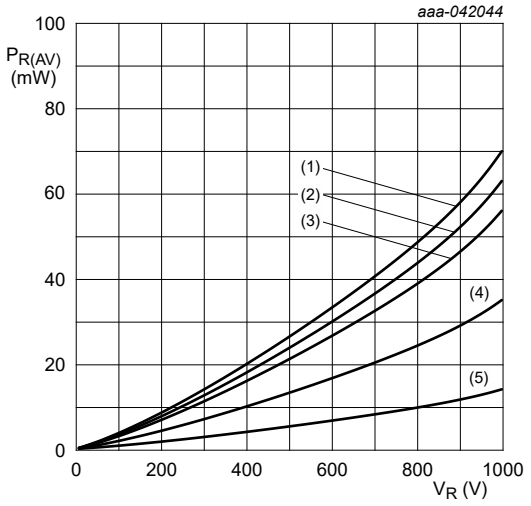
f = 1 MHz; T_{amb} = 25 °C

Fig. 5. Diode capacitance as a function of reverse voltage; typical values



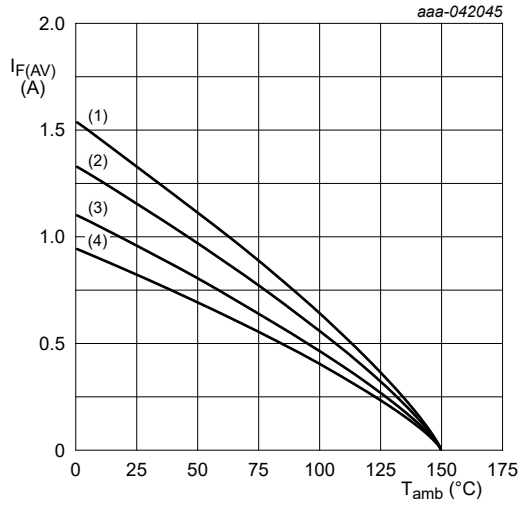
T_j = 150 °C
(1) δ = 0.1
(2) δ = 0.2
(3) δ = 0.5
(4) δ = 0.8
(5) δ = 1; DC

Fig. 6. Average forward power dissipation as a function of average forward current; typical values



T_j = 150 °C
(1) δ = 1; DC
(2) δ = 0.9
(3) δ = 0.8
(4) δ = 0.5
(5) δ = 0.2

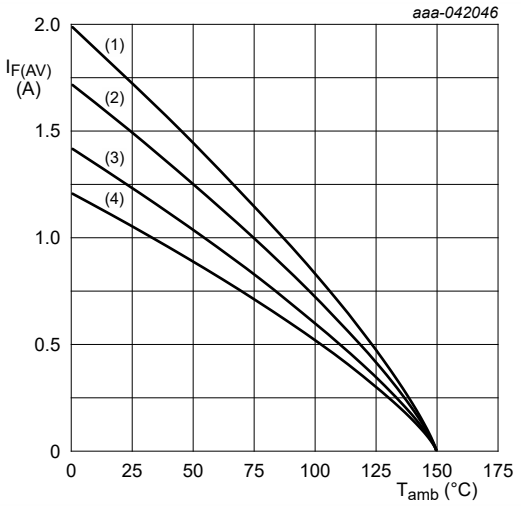
Fig. 7. Average reverse power dissipation as a function of reverse voltage; typical values



FR4 PCB, standard footprint
T_j = 150 °C
(1) δ = 1; DC
(2) δ = 0.5; f = 20 kHz
(3) δ = 0.2; f = 20 kHz
(4) δ = 0.1; f = 20 kHz

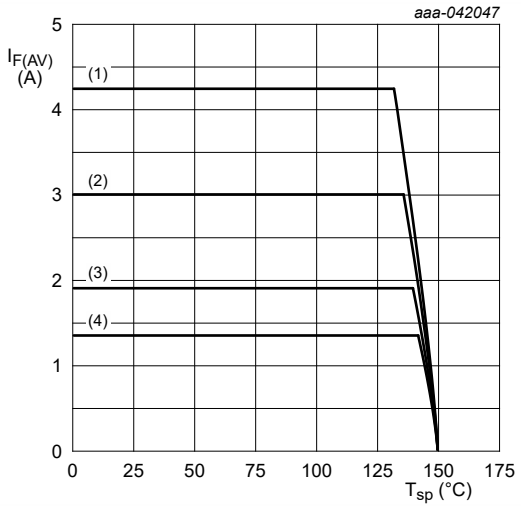
Fig. 8. Average forward current as a function of ambient temperature; typical values

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FR4 PCB, mounting pad for cathode 1 cm ^{2}
T_j = 150 °C
(1) δ = 1; DC
(2) δ = 0.5; f = 20 kHz
(3) δ = 0.2; f = 20 kHz
(4) δ = 0.1; f = 20 kHz

Fig. 9. Average forward current as a function of ambient temperature; typical values



T_j = 150 °C
(1) δ = 1; DC
(2) δ = 0.5; f = 20 kHz
(3) δ = 0.2; f = 20 kHz
(4) δ = 0.1; f = 20 kHz

Fig. 10. Average forward current as a function of solder point temperature; typical values

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11. Test information

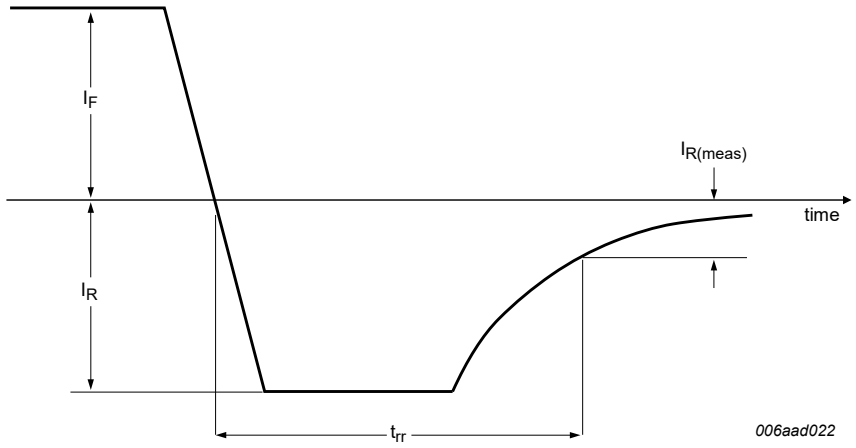


Fig. 11. Reverse recovery definition

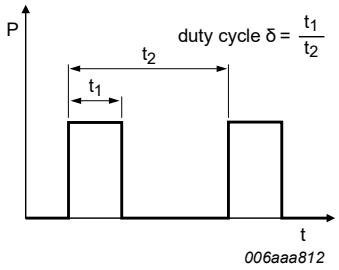


Fig. 12. Duty cycle definition

The current ratings for the typical waveforms are calculated according to the equations:

$I_{F(AV)} = I_M \times \delta$ with I_M defined as peak current,

$I_{RMS} = I_{F(AV)}$ at DC

$I_{RMS} = I_M \times \sqrt{\delta}$ with I_{RMS} defined as RMS current.

12. Package outline

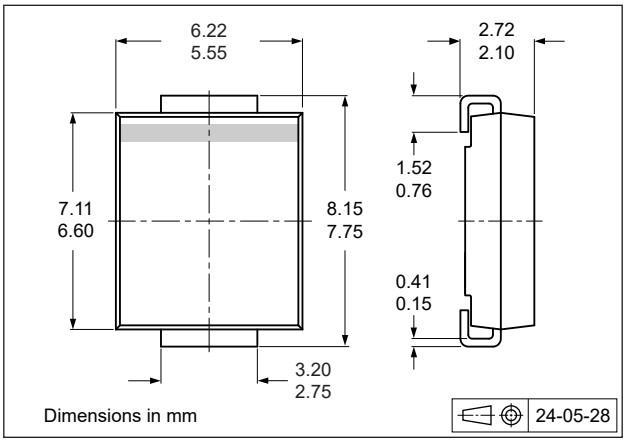


Fig. 13. Package outline SMC (SOD1003-1)

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13. Soldering

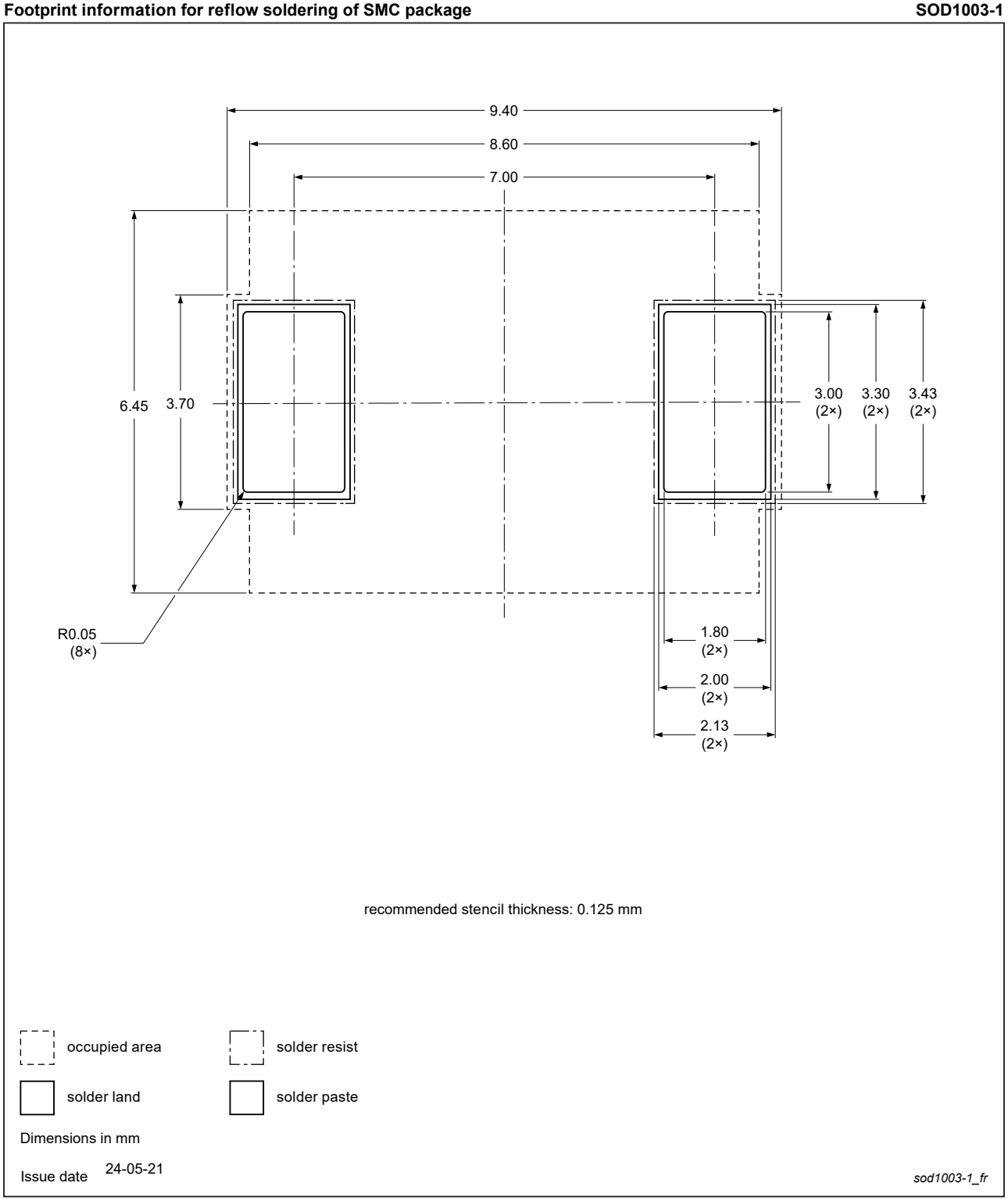


Fig. 14. Reflow soldering footprint for SMC (SOD1003-1)

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14. Revision history

Table 8. Revision history

Data sheet ID	Release date	Data sheet status	Change notice	Supersedes
US3M v.1	20250120	Product data sheet	-	-

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Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

[1] Please consult the most recently issued document before initiating or completing a design.
[2] The term 'short data sheet' is explained in section "Definitions".
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